

Developer Onboarding Guide

Project Title:

Mini Internal Developer Platform (Mini IDP)

Submitted By:

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Internship Organization:

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Project Domain:

T1 Cloud

1. Introduction

This guide helps developers understand how to use the Mini Internal Developer Platform for automated application deployment.

The platform automates:

- Docker image creation
- Infrastructure provisioning
- Amazon Web Services deployment
- Monitoring setup
- CI/CD workflow

2. Prerequisites

Before using the platform, ensure the following tools are installed:

Tool	Purpose
Git	Source code management
Node.js	Application runtime
Docker	Containerization
Terraform	Infrastructure provisioning
Amazon Web Services Account	Cloud deployment
GitHub Account	CI/CD integration

3. Clone the Repository

Clone the GitHub repository:

```
(base) sachin@SACHINs-MacBook-Air Desktop % git clone https://github.com/sachinbisen/Capg-Capstone-Project-Mini-Internal-Developer-Platform.git
```

Move into the project folder:

```
TERMINAL  ...  zsh - Capg-Capstone-Project-Mini-Internal-Developer-Platform  + v  [ ]  [ ]  ...  |  [ ]  [ ]  x

● (base) sachin@SACHINs-MacBook-Air Capg-Capstone-Project % cd Capg-Capstone-Project-Mini-Internal-Developer-Platform
(base) sachin@SACHINs-MacBook-Air Capg-Capstone-Project-Mini-Internal-Developer-Platform %
○ (base) sachin@SACHINs-MacBook-Air Capg-Capstone-Project-Mini-Internal-Developer-Platform %
```

4. Configure GitHub Secrets

Configure the following GitHub Secrets inside the repository:

Secret Name	Purpose
AWS_ACCESS_KEY_ID	Amazon Web Services authentication
AWS_SECRET_ACCESS_KEY	Amazon Web Services authentication
AWS_REGION	Amazon Web Services deployment region
DOCKER_USERNAME	Docker Hub authentication
DOCKER_PASSWORD	Docker Hub authentication

5. Local Application Setup

Install dependencies:

-> npm install

Run the application locally:

-> npm start

Application runs on:

-> http://localhost:3000

6. Docker Setup

Build Docker image locally:

-> docker build -t 260104/mini-idp-app:latest .

Run Docker container:

-> docker run -p 3000:3000 260104/mini-idp-app:latest

7. Terraform Setup

Move to Terraform directory:

-> cd terraform

Initialize Terraform:

-> terraform init

Validate Terraform configuration:

-> terraform validate

Apply infrastructure:

-> terraform apply -auto-approve

8. CI/CD Deployment Workflow

Deployment is fully automated using GitHub Actions.

Deployment Steps

1. Developer pushes code to GitHub
2. GitHub Actions pipeline starts automatically
3. Docker image gets built
4. Docker image is pushed to Docker Hub
5. Terraform provisions infrastructure
6. Amazon EC2 deploys latest application container
7. Amazon CloudWatch monitoring becomes active

9. Monitoring Access

CloudWatch monitoring can be accessed from:

-> AWS Console → CloudWatch

Available monitoring features:

- CPU utilization monitoring
- EC2 status monitoring
- Network monitoring
- CloudWatch dashboard visualization

10. Terraform Remote State

Terraform state is stored remotely using AWS S3.

S3 Backend Bucket:

-> sachin-mini-idp-terraform-state

Benefits:

- Prevents duplicate infrastructure
- Centralized state management
- Better CI/CD reliability

11. Deployment Verification

After deployment, verify:

Application Access

-> `http://EC2_PUBLIC_IP:3000`

CloudWatch Dashboard

-> AWS Console → CloudWatch → Dashboards

GitHub Actions Pipeline

-> GitHub Repository → Actions

12. Troubleshooting

Problem	Solution
Terraform validation error	Run “ <code>terraform fmt</code> ” and “ <code>terraform validate</code> ”
Docker build failure	Verify Docker installation
GitHub Actions failure	Check GitHub Secrets configuration
EC2 deployment issue	Verify AWS region and permissions
CloudWatch metrics missing	Wait for metrics collection from Amazon CloudWatch

13. Conclusion

The Mini Internal Developer Platform provides a simplified DevOps automation workflow for developers.

The platform demonstrates practical implementation of:

- CI/CD automation using GitHub Actions
- Infrastructure as Code using Terraform
- Docker containerization
- Amazon Web Services cloud deployment
- Cloud monitoring and observability using Amazon CloudWatch

The platform helps developers deploy applications faster with reduced manual effort and improved deployment consistency.